

FRAUD 101

COURSE COMPANION GUIDE

Fraud 101

A practical reference for connected fraud analysis, coordinated attacks, identity and payment controls, decisioning, investigations, performance measurement, and incident response.

Use this guide to:

Review course concepts, structure an investigation, plan a control test, prepare an incident response, and document your next actions.

COURSE COMPANION GUIDE

HOW TO USE THIS GUIDE

Turn Course Concepts into Repeatable Practice

The guide follows the six instructional lessons and ends with reusable worksheets. It is a job aid, not a substitute for your organization's policies, contracts, legal advice, or payment-network rules.



During the course

Record examples from your environment. Compare the course framework with your current tools, data, decision rights, and customer journey.



After the course

Use the worksheets in fraud reviews, control-design meetings, investigation handoffs, incident exercises, and post-incident reviews.

Course map

1	Understanding Advanced Fraud Risk	Connect events across time, systems, and entities.
2	Coordinated Attacks, Automation, and AI-Enabled Fraud	Recognize preparation, execution, monetization, and adaptation.
3	Identity, Account Security, and Payment Authentication	Separate identity, recovery, fraud screening, authentication, and authorization.
4	Fraud Decisioning, Models, and Investigations	Test decisions, investigate relationships, and govern analytical tools.
5	First-Party Misuse, Policy Abuse, and Fraud Performance	Classify loss accurately and measure business trade-offs.
6	Fraud Operations, Incident Response, and Capstone	Contain harm, preserve evidence, coordinate owners, and improve controls.

Core principle: a single event or signal rarely tells the full story. Connect the customer journey, test a clear hypothesis, and choose a proportionate response.

LESSON 1

Understanding Advanced Fraud Risk

Advanced analysis connects account, payment, fulfillment, support, refund, and dispute activity instead of reviewing each event in isolation.

Four levels of connected fraud

**Fraud event**

One suspicious login, order, refund, account change, or payout request.

**Fraud pattern**

Repeated behavior that shares attributes across events or entities.

**Fraud campaign**

A coordinated series targeting a product, promotion, process, or vulnerability.

**Fraud network**

Linked accounts, devices, identities, payments, addresses, or beneficiaries working together.

Build a testable fraud hypothesis

- Observation: state what changed and the period in which it changed.
- Hypothesis: explain the suspected behavior and how it may create loss.
- Supporting evidence: list signals that increase confidence.
- Contradicting evidence: list facts that could support a legitimate explanation.
- Required data: identify what would confirm, narrow, or disprove the hypothesis.
- Customer impact: estimate who could be delayed, blocked, or challenged.

A good hypothesis can be tested and disproved. It should not be a conclusion written before the investigation begins.

LESSON 1 WORKSHEET

Connected Fraud Hypothesis

What changed?	
Where in the journey?	
Suspected behavior	
Supporting signals	
Contradicting signals	
Data still needed	
Potential exposure	
Potential customer impact	
Decision and owner	

Relationship-strength check

- How distinctive is the shared attribute?
- Did the links appear in a meaningful time window?
- How dense is the relationship among the entities?
- Is there a plausible legitimate business explanation?
- Does value move to the same destination or beneficiary?

LESSON 2

Coordinated Attacks, Automation, and AI-Enabled Fraud

AI and automation can make established attacks faster, cheaper, more personalized, and more convincing. Detection should focus on behavior, relationships, and verification discipline.

Attack lifecycle



1. Reconnaissance

Research products, controls, support procedures, limits, and fulfillment.



2. Resource acquisition

Obtain credentials, identities, proxies, bots, mules, and resale services.



3. Testing and execution

Probe controls, then increase value, volume, or complexity.



4. Monetization

Extract value through goods, stored value, refunds, payouts, or resale.



5. Concealment and adaptation

Change infrastructure, identities, behavior, products, or timing.



Analyst focus

Follow both the purchase and the destination of value.

Multi-dimensional velocity

- Measure activity across identities, devices, sessions, payment methods, addresses, beneficiaries, and time.
- Look for the same sequence repeated across distributed infrastructure.
- Separate weak shared infrastructure from stronger links such as reused payments, destinations, or synchronized actions.

Writing quality, realistic images, convincing audio, or a familiar label do not establish legitimacy. Verify high-risk actions through trusted channels and account history.

LESSON 2 WORKSHEET

Coordinated Attack Assessment

Observed stage	Reconnaissance / acquisition / execution / monetization / adaptation
Entities involved	
Shared attributes	
Repeated sequence	
Velocity dimensions	
Value destination	
Legitimate explanation	
Control response	
Evidence to preserve	

Layered response options



Reduce opportunity

Rate-limit repeated failures, narrow sensitive actions, and require stronger verification for material changes.



Protect legitimate customers

Target the affected journey or relationship. Avoid broad blocks when a narrower control can contain the risk.

LESSON 3

Identity, Account Security, and Payment Authentication

Identity and payment controls answer different questions. A successful payment decision does not prove that account recovery, customer intent, or the complete purchase journey was legitimate.

Account authentication	Can the user present an accepted authenticator?	That recovery or recent account changes were legitimate.
Merchant fraud screening	Does the transaction fit the merchant's risk policy?	That the cardholder personally approved the purchase.
Payment authentication	Was the cardholder or payment user authenticated?	That the order, delivery, or account journey is free of fraud.
Issuer authorization	Will the issuer approve the payment request?	That the merchant should release goods or remove a risk hold.

High-risk recovery controls

- Verify through an established contact method or trusted channel.
- Notify the established contact when recovery details change.
- Temporarily restrict payout, delivery, credential, or profile changes.
- Review linked sessions, devices, network context, and recent account history.
- Remove restrictions only after the recovery evidence and follow-on activity are resolved.

Passkeys can strengthen authentication, but recovery and support processes still need independent controls.

LESSON 3 WORKSHEET

Recovery and Sensitive-Action Decision

Triggering event	
Authentication evidence	
Recovery evidence	
Established-contact check	
Recent sensitive changes	
Payment decision	
Merchant risk decision	
Temporary restriction	
Release criteria	
Decision owner	

LESSON 4

Fraud Decisioning, Models, and Investigations

Strong fraud decisioning combines reliable data, understandable rules or models, safe testing, investigation discipline, and accountable owners.

A controlled change cycle



1. Define

State the problem, target population, expected benefit, risks, and measures.



2. Test

Use shadow mode, a holdout, staged rollout, or another safe comparison.



3. Measure

Track fraud, approval, customer friction, review workload, and displacement.



4. Decide

Scale, revise, pause, or stop using predefined thresholds and accountable approval.

Investigation structure

- Begin with the event, then build outward to accounts, devices, payments, addresses, beneficiaries, and behavior.
- Record why each relationship matters and what legitimate explanation remains possible.
- Separate observed facts, analytical inferences, confidence, and final decisions.
- Preserve supporting and contradicting evidence, not only the evidence that supports the final action.

A manual-review decline is a decision outcome, not automatically a confirmed-fraud label. Weak labels can weaken future rules and models.

LESSON 4 WORKSHEET

Control Test and Investigation Record

Control test

Problem and population	
Proposed control	
Test method	
Success measures	
Customer safeguards	
Rollback threshold	
Owner and review date	

Investigation handoff

Hypothesis and confidence	
Key linked entities	
Exposure estimate	
Contradicting evidence	
Decision and rationale	
Follow-up actions	

LESSON 5

First-Party Misuse, Policy Abuse, and Fraud Performance

Accurate classification supports fair customer treatment, better controls, and trustworthy reporting. Similar losses may require different evidence and remedies.

Classify before you optimize



Third-party fraud

An unauthorized actor uses another person's account, identity, or payment method.



First-party misuse

The customer participates in the transaction but later misrepresents intent, receipt, or responsibility.



Policy abuse

A person exploits promotion, return, refund, or service rules without necessarily using a stolen identity.



Legitimate dispute

The customer has a genuine product, delivery, billing, service, or authorization concern.

Balanced fraud performance

- Loss: confirmed fraud, abuse, refunds, disputes, and recovery.
- Approval: accepted transactions and legitimate revenue.
- Customer experience: friction, abandonment, complaints, and time to resolution.
- Operations: review volume, handling time, escalations, and backlog.
- Displacement: movement into another product, channel, refund path, or loss category.

A lower reported fraud rate can hide more false positives, customer friction, manual work, or displacement. Read the measures together.

LESSON 5 WORKSHEET

Balanced Control Scorecard

Fraud loss			
Approval / revenue			
Customer friction			
Operational workload			
Displacement			
Segment impact			

Decision review

What improved?	
What worsened?	
Which segments were affected?	
Where did activity move?	
Scale / revise / pause / stop	

LESSON 6

Fraud Operations and Incident Response

An incident response should contain harm, preserve evidence, coordinate decision owners, communicate consistently, and produce durable improvements.

The first minutes of a fraud incident



1. Confirm and contain

Validate the active pattern. Pause risky transactions or fulfillment, disable exposed credentials, restrict compromised destinations, and protect affected accounts.



2. Escalate and assign

Name one incident lead. Bring in fraud, security, payments, service, legal, communications, and leadership as needed.



3. Preserve and assess

Retain alerts, logs, decisions, communications, affected entities, timestamps, and estimated exposure.



4. Recover and monitor

Restore safely, notify the appropriate parties, monitor for displacement, and remove emergency controls deliberately.

Use the narrowest effective containment

- Target the affected journey, capability, destination, or relationship.
- Define what the emergency control stops and who can change it.
- Set an expiration or review time so temporary controls do not become permanent by accident.
- Monitor for adaptation into a different account, payment, product, support, or refund path.

Preserve evidence while the systems and decisions are still observable. Recovery without an evidence trail weakens investigation, reporting, and future prevention.

LESSON 6 WORKSHEET

Incident Response Record

Incident lead	
Start time and trigger	
Affected journeys	
Known entities	
Estimated exposure	
Immediate containment	
Evidence preserved	
Teams and partners notified	
Customer action	
Recovery criteria	
Monitoring period	
Next update time	

POST-INCIDENT REVIEW

Turn the Incident into Durable Improvement

Use a blameless review that identifies decisions, control ownership, customer impact, and evidence gaps.

What happened?	
How was it detected?	
Which control worked?	
Which control failed or was missing?	
What delayed containment?	
What customer harm occurred?	
What activity was displaced?	
Permanent change	
Monitoring and rollback	
Owner and due date	
Effectiveness review date	

COURSE REVIEW

Advanced Fraud Readiness Checklist

- [] We connect account, payment, fulfillment, support, refund, and dispute evidence.
- [] We distinguish events, patterns, campaigns, and networks.
- [] We use testable hypotheses with supporting and contradicting evidence.
- [] We measure velocity across entities, devices, sessions, payments, destinations, and time.
- [] We verify sensitive recovery actions through trusted channels.
- [] We separate authentication, fraud screening, payment authentication, and authorization.
- [] We test new rules or models safely and define rollback criteria.
- [] We preserve investigation evidence and record confidence and rationale.
- [] We distinguish third-party fraud, first-party misuse, policy abuse, and legitimate disputes.
- [] We measure fraud, approval, customer experience, workload, and displacement together.
- [] We have named incident owners, containment authority, and update routines.
- [] We conduct post-incident reviews with owners and effectiveness dates.

My next three actions

1	
2	
3	

REFERENCES

Primary Resources and Further Reading

Use current versions of standards, network rules, contracts, and organizational policies. External resources may change after publication.

- [1] Fraud 101. Course curriculum and course resources.
- [2] Association for Financial Professionals. 2026 Payments Fraud and Control Survey Report: Key Highlights.
- [3] OWASP Foundation. Automated Threats to Web Applications.
<https://owasp.org/www-project-automated-threats-to-web-applications/>
- [4] National Institute of Standards and Technology. Digital Identity Guidelines: Authentication and Authenticator Management, SP 800-63B. <https://pages.nist.gov/800-63-4/sp800-63b.html>
- [5] FIDO Alliance. Passkeys. <https://fidoalliance.org/passkeys/>
- [6] EMVCo. EMV 3-D Secure. <https://www.emvco.com/emv-technologies/3-d-secure/>
- [7] PCI Security Standards Council. Document Library. https://www.pcisecuritystandards.org/document_library/

This guide supports learning and operational planning. It does not replace legal advice, contractual obligations, payment-network rules, or your organization's approved policies and incident procedures.